

When Does Adaptive Coordinate Sampling Pay for Itself?

Ridge-Regression Baselines, ASCD, and Oracle-Quality Analysis

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Abstract

We study coordinate-selection rules for ridge regression under feature scaling and correlation. A baseline experiment compares uniform sampling, fixed Lipschitz sampling, periodically refreshed gradient-gain sampling, and a progress-triggered gradient-gain heuristic across four synthetic regimes and the packaged diabetes regression dataset. Each method is evaluated over 12 seeded runs using a common coordinate update and a 2,400-update budget.

We subsequently implement Approximate Steepest Coordinate Descent (ASCD) of Stich, Raj, and Jaggi using the paper’s zero oracle with exact full-gradient initialization. On the scaled-correlated case, this configuration reaches the target in 9 of 12 runs compared with 6 of 12 for uniform sampling, but records greater proxy work on all five seeds where both methods succeed.

A further oracle-quality analysis explains why the zero-oracle ASCD active set remains effectively full. On the synthetic cases the full active set is used on approximately 99.94% of updates, and the median fraction of coordinates with zero certified lower bound is one. An exact inner-product oracle reduces the median active set to one coordinate and reaches the target in all runs. After charging a modelled dense-Gram setup cost, the exact oracle is cheaper than the zero oracle on the correlated cases and diabetes but more expensive on the two independent synthetic cases. Simulated approximate oracles at the tested intermediate precisions do not provide a useful middle regime. These experiments are comparisons on our frozen benchmark rather than reproductions of the paper’s published figures, and no claim of general superiority or algorithmic novelty is made.

1 Motivation and related work

Randomized updates and adaptive information collection feature in stochastic block BFGS [1], minibatch semi-stochastic optimization [2], parallel and accelerated coordinate descent [3, 5], and randomized linear solvers [4]. Adaptive coordinate-selection methods include adaptive probabilities for stochastic dual coordinate ascent [6], adaptive importance sampling [7], and Approximate Steepest Coordinate Descent (ASCD) [8].

These works prevent gradient-informed sampling or refreshes alone from being treated as a new idea. The practical question considered here is instead when additional coordinate-selection information improves optimization enough to compensate for the work required to obtain and maintain it.

The study proceeds in three stages. First, four baseline selection rules are compared. Second, a published ASCD configuration using the paper’s zero oracle is implemented on the same frozen benchmark. Third, the ASCD active-set mechanism is diagnosed and compared across zero, approximate, and exact oracle-quality configurations.

2 Experimental setup and baseline methods

For a design matrix $X \in \mathbb{R}^{n \times d}$ and response y , we minimize

$$f(w) = \frac{1}{2} \|Xw - y\|_2^2 + \frac{\lambda}{2} \|w\|_2^2, \quad \lambda = 0.03. \quad (1)$$

With residual $r = Xw - y$,

$$g_j = X_j^\top r + \lambda w_j, \quad L_j = \|X_j\|_2^2 + \lambda, \quad (2)$$

and each baseline method applies

$$w_j \leftarrow w_j - \frac{g_j}{L_j} \quad (3)$$

and updates the residual.

Uniform sampling uses $p_j = 1/d$, whereas fixed Lipschitz sampling uses

$$p_j = \frac{L_j}{\sum_k L_k}.$$

Both gradient-gain variants use

$$p_j = \frac{0.05}{d} + 0.95 \frac{g_j^2/L_j}{\sum_k g_k^2/L_k}, \quad (4)$$

with uniform fallback when the denominator vanishes.

Scheduled gain refreshes initially and every 60 updates. Triggered gain refreshes at step zero and subsequently at a 120-step checkpoint if the latest objective decrease is below half the preceding decrease, subject to a 60-step minimum interval.

The four synthetic cases cross balanced versus geometrically scaled features with independent versus eight-group correlated features, with $n = 400$ and $d = 48$. The packaged diabetes design provides a fifth case.

Each method uses 12 problem seeds and at most 2,400 coordinate updates. A dense ridge solve defines f^* for evaluation. A run succeeds if a saved checkpoint satisfies

$$f(w) - f^* \leq 10^{-4}.$$

The reference optimum is used only for evaluation and not by the selection rules.

3 Baseline results

Table 1: Baseline runs reaching the target within the 2,400-update budget (out of 12).

Case	Uniform	Lipschitz	Scheduled	Triggered
Balanced independent	12	12	12	12
Scaled independent	12	0	12	12
Balanced correlated	12	12	0	0
Scaled correlated	6	0	0	0
Diabetes	12	12	9	5

Uniform and fixed Lipschitz sampling both succeed in all runs of the balanced-independent, balanced-correlated, and diabetes cases. Lipschitz sampling fails on the scaled-independent case, where the other three baseline methods succeed. The two gradient-gain heuristics fail to reach the target in either correlated synthetic case within the update budget.

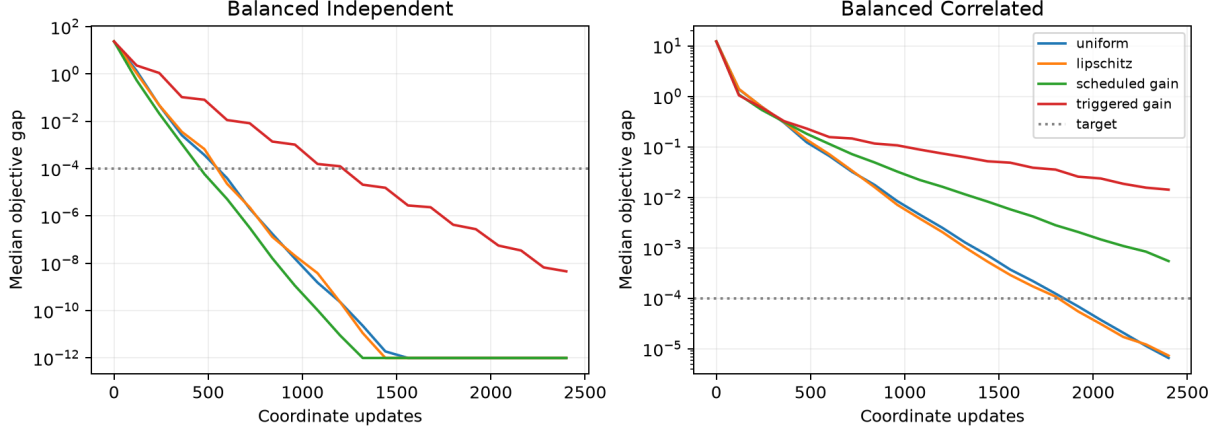


Figure 1: Median objective gaps for the two balanced baseline cases. The horizontal dotted line marks the 10^{-4} target. A median trajectory does not display variability across individual runs.

4 Published ASCD comparison

4.1 Selection rule

ASCD [8] maintains, for each coordinate j , an estimated gradient $\tilde{g}_j$ and certified radius r_j satisfying

$$|g_j - \tilde{g}_j| \leq r_j.$$

The corresponding upper and lower bounds are

$$u_j = |\tilde{g}_j| + r_j, \quad \ell_j = \max\{0, |\tilde{g}_j| - r_j\}. \quad (5)$$

The active set is the smallest nonempty subset I such that every excluded coordinate satisfies

$$u_j^2 < \frac{1}{|I|} \sum_{i \in I} \ell_i^2. \quad (6)$$

Equality therefore does not permit exclusion. A coordinate is drawn uniformly from members of I attaining the largest lower bound.

Our implementation initializes $\tilde{g}$ with the exact full gradient and all radii at zero.

4.2 Zero oracle

For the Stage 3 comparison, the paper’s zero cross-coordinate oracle is used. If coordinate i changes by h , then for $j \neq i$

$$g_j^{\text{new}} - g_j = h X_j^\top X_i. \quad (7)$$

Cauchy–Schwarz gives

$$|g_j^{\text{new}} - g_j| \leq |h| \|X_j\|_2 \|X_i\|_2. \quad (8)$$

The stored cross-gradient estimate is therefore unchanged and its certified radius is increased by this bound.

For the selected coordinate,

$$h = -\frac{g_i}{L_i},$$

and exact minimization gives

$$g_i^{\text{new}} = g_i + h L_i = 0.$$

The selected estimate and radius can therefore be reset to zero up to numerical precision.

Table 2: Uniform and zero-oracle ASCD results on the frozen benchmark. Work to target is the median among successful runs under each method’s recorded proxy.

Case	Uniform		ASCD		
	Success	Work	Success	Work	Median $ I $
Balanced independent	12/12	513,288	12/12	762,936	48
Scaled independent	12/12	513,288	12/12	762,936	48
Balanced correlated	12/12	1,641,536	12/12	2,249,416	48
Scaled correlated	6/12	1,897,956	9/12	2,844,008	48
Diabetes	12/12	919,234	12/12	984,864	10

4.3 Initial ASCD results

The scaled-correlated case motivates a paired comparison because the successful-run medians use different seed sets.

The two methods both reach the target on seeds 0, 4, 5, 6, and 10. On those five shared successes, median recorded work to target is 1,846,672 for uniform and 2,844,008 for ASCD. The median paired ASCD-minus-uniform difference is 997,336 proxy units and the median recorded-work ratio is 1.5401. ASCD records greater proxy work on all five shared successes.

These proxy values are descriptive rather than exact FLOP or wall-clock comparisons because the two implementations do not have fully harmonized setup boundaries.

5 Stage 4: oracle-quality mechanism analysis

5.1 Zero-oracle diagnosis

The Stage 3 median active set alone does not reveal whether the active set is usually large or merely has a large median. Stage 4 therefore replays the frozen zero-oracle trajectory while recording per-update bound and active-set diagnostics.

The replay validates every saved Stage 3 checkpoint and reproduces the frozen objective gaps and proxy work.

For the four synthetic cases, the full 48-coordinate active set is used on approximately 99.94% of recorded updates. On diabetes, the full 10-coordinate set is used on approximately 99.96% of updates. The median fraction of coordinates with zero lower bound is 1.0000 for every case.

Since

$$\ell_j = \max\{0, |\tilde{g}_j| - r_j\},$$

a zero lower bound occurs when the certified radius dominates the stored estimate. If all lower bounds are zero, then

$$\frac{1}{|I|} \sum_{i \in I} \ell_i^2 = 0,$$

so no coordinate can satisfy the strict safe-exclusion condition $u_j^2 < 0$.

The diagnostic radius trajectories are consistent with this mechanism: the certified radius generally remains above the typical true-gradient scale.

5.2 Oracle-quality experiment

Phase B retains the frozen benchmark and coordinate update while changing the oracle information.

We compare:

- the zero oracle g^3 ;

- simulated approximate oracles g^2 with $\epsilon \in \{1, 0.5, 0.25, 0.125\}$;
- the exact inner-product oracle g^1 .

The simulated g^2 runs isolate oracle-quality effects only. They do not implement or cost a practical embedding or approximation-acquisition mechanism.

The g^3 configuration reproduces the frozen Stage 3 trajectory.

Table 3: Phase B runs reaching the target within 2,400 updates (out of 12).

Case	g^3	$g^2(1)$	$g^2(0.5)$	$g^2(0.25)$	$g^2(0.125)$	g^1
Balanced independent	12	12	0	0	0	12
Scaled independent	12	12	0	0	0	12
Balanced correlated	12	12	0	0	0	12
Scaled correlated	9	11	0	0	0	12
Diabetes	12	12	0	0	0	12

5.3 Active-set transition

The exact oracle produces a qualitatively different selection regime.

For g^1 , the median active set is one coordinate on every case, and the 10th, 25th, 50th, 75th, and 90th active-set percentiles are all one. In contrast, g^3 and simulated g^2 with $\epsilon = 1$ retain the full coordinate dimension as their median active set.

The intermediate simulated g^2 configurations technically prune on almost every update, but only weakly: their synthetic median active set is 47 of 48 coordinates, and the diabetes median is 9 of 10 coordinates.

Thus the tested approximate precision levels do not provide a useful middle regime between the zero and exact oracles.

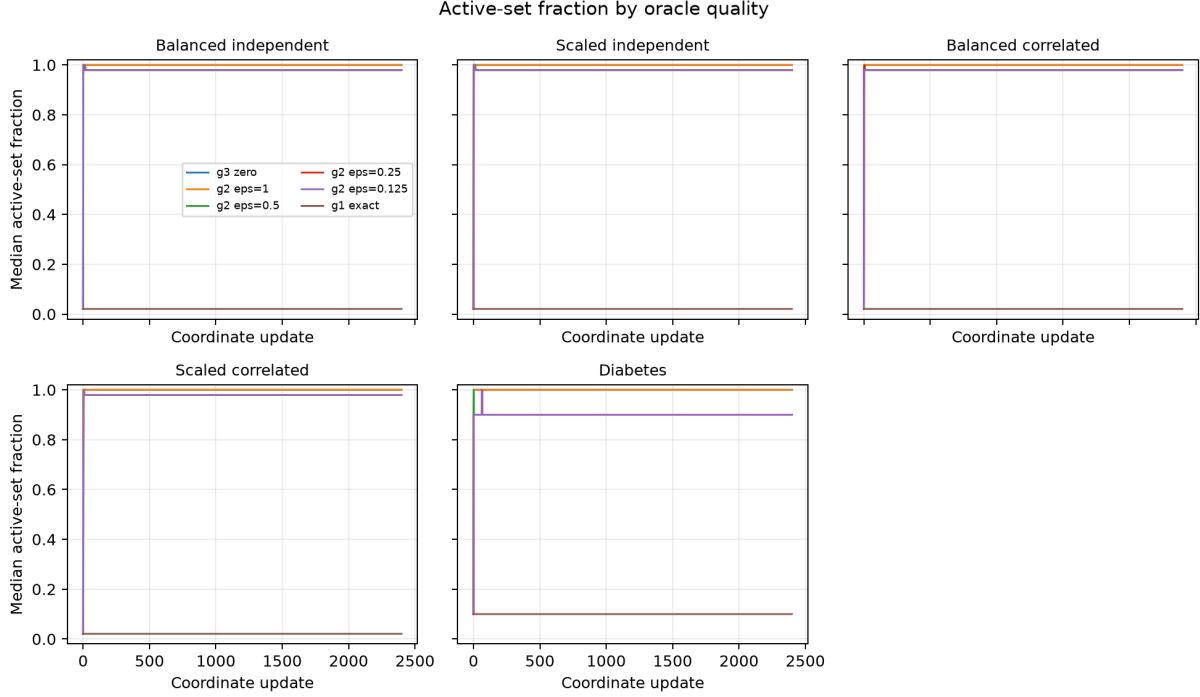


Figure 2: Median active-set fraction across the Phase B oracle-quality configurations. Exact g^1 produces an effectively one-coordinate active set, whereas the tested nonzero-error configurations remain close to the full coordinate set.

5.4 Optimization behavior

The objective-gap trajectories show the same qualitative separation.

Exact g^1 reaches the target in every run and converges substantially faster in coordinate updates. The simulated $\epsilon = 1$ configuration largely resembles the zero oracle. The intermediate simulated configurations with $\epsilon \in \{0.5, 0.25, 0.125\}$ do not reach the target in any run.

This non-monotonic behavior should be interpreted narrowly. Changing the approximate oracle changes both the maintained gradient estimates and the coordinate-selection trajectory. The experiment therefore does not show that tighter approximate oracles are generally harmful; it shows that the tested simulated configurations do not produce a successful intermediate regime under this benchmark.

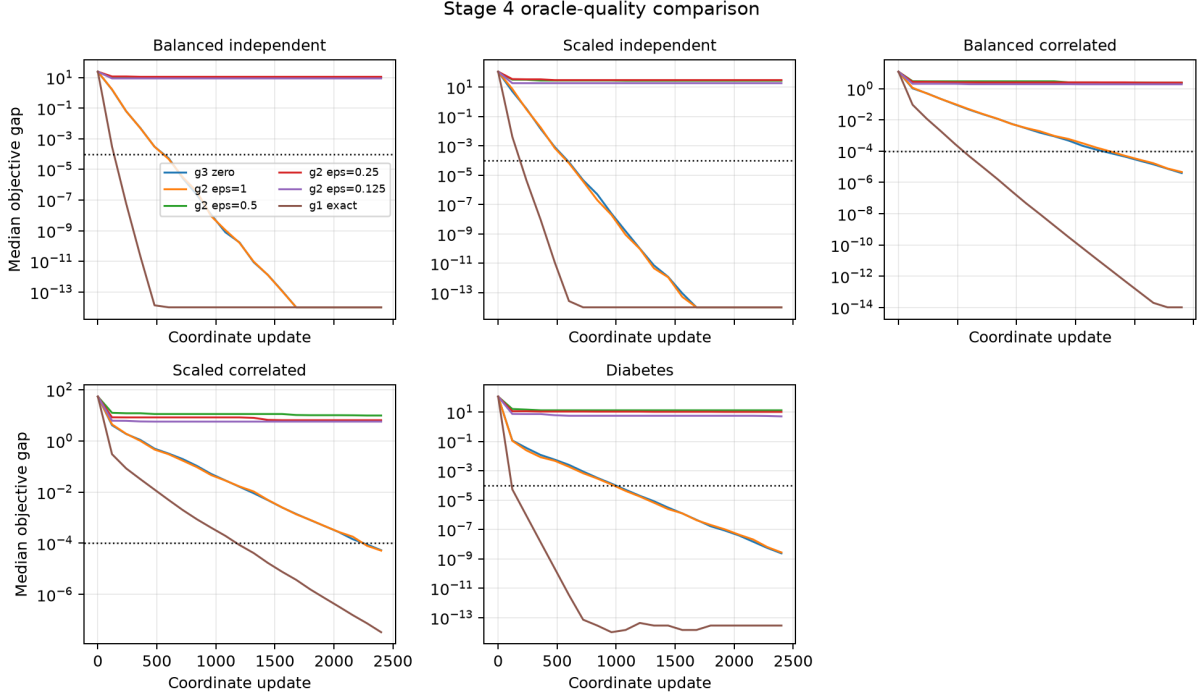


Figure 3: Median objective-gap trajectories for the Phase B oracle-quality comparison. The dotted line marks the 10^{-4} target.

5.5 Does exact oracle information pay for itself?

The common ASCD core proxy does not include the cost of obtaining an exact Gram matrix. For g^1 , Stage 4 therefore adds a separate dense setup charge

$$nd^2.$$

For the synthetic cases this is

$$400 \cdot 48^2 = 921,600,$$

and for diabetes it is

$$442 \cdot 10^2 = 44,200.$$

Table 4: Costed zero-oracle and exact-oracle comparison. The final column is the median paired g^1/g^3 recorded-work ratio on shared successful seeds.

Case	g^3 success	g^1 success	g^3 work	g^1 work	Paired ratio
Balanced independent	12/12	12/12	762,936	1,238,592	1.6235
Scaled independent	12/12	12/12	762,936	1,238,592	1.6235
Balanced correlated	12/12	12/12	2,249,416	1,684,536	0.7489
Scaled correlated	9/12	12/12	2,844,008	2,427,776	0.8536
Diabetes	12/12	12/12	984,864	164,374	0.1675

The exact oracle therefore does not have a uniform cost advantage.

It is more expensive under the modelled proxy on the two independent synthetic cases. It is cheaper on balanced-correlated, cheaper on the shared scaled-correlated successes while also succeeding on three additional seeds, and substantially cheaper on diabetes.

These are accounting-model results rather than measured FLOP or wall-clock ratios.

6 Interpretation and limitations

The baseline experiment shows that the simple gradient-gain heuristics do not succeed consistently across the frozen cases.

The initial zero-oracle ASCD comparison improves scaled-correlated success count under a fixed update budget, but does not show lower recorded work on shared successful seeds.

Stage 4 explains a major part of this behavior. Under the zero oracle, certified uncertainty overwhelms the stored estimates sufficiently often that almost every lower bound becomes zero. The safe active set is therefore effectively full.

Exact oracle information reverses this mechanism completely and produces an active set of one coordinate. This improves optimization strongly, but the cost of obtaining the exact information determines whether the overall proxy improves.

The result is regime-dependent rather than universal.

The simulated approximate-oracle experiment has an important limitation: it measures oracle quality without implementing a realistic acquisition mechanism. Its results therefore cannot answer whether a practical approximate inner-product oracle would pay for itself.

The tested intermediate precision settings also produce an unusual non-monotonic optimization pattern. They should be reported as a result of this particular simulation and benchmark rather than generalized.

Only one small packaged real dataset is included. No predictive generalization performance is evaluated. The proxy is not an exact FLOP count, and wall-clock measurements depend on software, hardware, and setup boundaries.

Finally, the ASCD paper’s one-step comparison assumes common smoothness and update conditions that are not identical to this benchmark, which uses coordinate-specific L_j values and its existing exact coordinate-minimization rule. The paper’s theoretical result should therefore not be interpreted as a guarantee for the experiments reported here.

7 Conclusion and future research

The study now provides a progression from exploratory baseline sampling rules to a published ASCD mechanism and then to a direct analysis of oracle quality.

The zero-oracle ASCD active set remains almost always full because its certified lower bounds collapse to zero. Exact inner-product information removes this mechanism, reduces the median active set to one coordinate, and reaches the target in every run.

Whether the extra information pays for itself depends on the benchmark regime. Under the modelled dense-Gram setup charge, exact g^1 is more expensive than g^3 on the independent synthetic cases but cheaper on the correlated cases and diabetes.

The tested simulated approximate oracles do not provide a successful middle regime. Under the predeclared Stage 4 decision rule, these results do not justify proceeding directly to a costed implementation of the tested approximate precision levels.

A finer precision sweep between $\epsilon = 0.125$ and the exact limit could be investigated later, but would be a post-hoc exploratory extension. Likewise, a genuinely costed approximate-inner-product implementation would require its acquisition mechanism to be specified and charged explicitly before making cost-effectiveness claims.

Broader datasets, repeated wall-clock measurements, uncertainty analyses, and harmonized implementation boundaries would be required before making stronger claims about general computational advantage.

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oracle-comparison workflows, plotting, analysis, and report text. The code and generated outputs accompany the report for reproducibility. Descriptions of independent authorship should reflect the author’s actual personal verification, decisions, and revisions. Applicable citation and AI-disclosure requirements should be followed.

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